

3.1 Conversion of Decimals to Rational Numbers.

We have already learnt that decimals are those numbers which contain digits whose value is in tenths, hundredths, thousandths and so on.

Let us consider a decimal number 2.5, it comprises of two parts: one '2' which is a whole part and another is '.5' which is a decimal part.

It has one place of decimal because there is only one digit after decimal point.

A decimal may have whole part and decimal part.

as: 2.5, 14.57, 32.163 and 103.0978 etc.

or a decimal may have a decimal part only

as: 0.6, 0.56, 0.10652 and so on.

3.1.1 Convert decimals to rational numbers

We can convert decimals to rational numbers as follows:

Example 1: Convert a decimal 0.5 to rational number.

$$\begin{aligned} \text{Solution: } 0.5 &= \frac{05}{10} = \frac{5}{2 \times 5} = \frac{\overset{1}{\cancel{5}}}{2 \times \cancel{5}_1} = \frac{1}{2} \\ \text{or } 0.5 &= \frac{\overset{1}{\cancel{5}}}{\cancel{10}_2} = \frac{1}{2} \end{aligned}$$

Steps:

- (i) Write 1 below the decimal point and remove decimal point.
- (ii) Put as many zeros after 1 as the number of digits after decimal point. (As it has one place of decimal, so we put one zero after 1).
- (iii) Reduce it to the lowest form (Standard form)

Thus decimal 0.5 = $\frac{1}{2}$ (Rational number).

Example 2: Convert $\frac{1}{2}$ decimal 0.06 to rational number.

Solution: As 0.06 has two place of decimal, so we put two zeros after '1'.

$$0.06 = \frac{06}{100} = \frac{\overset{3}{\cancel{6}}}{\cancel{100}_{50}} = \frac{3}{50}$$

Thus, we get decimal $0.6 = \frac{3}{50}$ (Rational number).

Hence $\frac{1}{2}$ and $\frac{3}{50}$ are rational numbers.

Example 3: Convert decimal 2.40 into rational number.

Solution: Given decimal 2.40 comprises of two parts '2' a whole part and '.40' a decimal part.

So, we first convert decimal part '.40' into rational number.

$40 = \frac{40}{100}$ (As .40 has two places of decimal, so we put two zeros after '1').

$$\text{So, } 2.40 = 2 + .40 = 2 \frac{40}{100} = 2 \frac{4}{10} = 2 \frac{2}{5} = \frac{12}{5}$$

Thus decimal $2.40 = \frac{12}{5}$ which is a common fraction or a rational number.

Example 4. Convert - 1.60 into rational number.

Here whole part '-1' is kept as a whole and the decimal part '.60' is to be converted into common fraction.

$$\begin{aligned} \text{So, } -1.60 &= -(1 + .60) \\ &= -\left(1 + \frac{60}{100}\right) = -\left(1 + \frac{6}{10}\right) \\ &= -\left(1 + \frac{3}{5}\right) = -1\frac{3}{5} = -\frac{8}{5} \end{aligned}$$

Thus decimal $-1.60 = -\frac{8}{5}$ which is a rational number.

EXERCISE 3.1

A. Convert the following one place-decimals into rational numbers.

- (1) 0.2 (2) 0.4 (3) 0.6 (4) 0.8 (5) 0.9

B. Convert the following two place-decimals into rational numbers.

- (1) 0.15 (2) 0.35 (3) 0.48 (4) 0.75 (5) 0.95
(6) - 1.30 (7) - 2.50 (8) 5.05 (9) 5.60 (10) 6.85

C. Convert the following three place-decimals into rational numbers.

- (1) 0.024 (2) 3.125 (3) 0.375 (4) 4.065 (5) 5.625
 (6) -0.001 (7) -0.0005 (8) -0.005 (9) -5.005
 (10) -11.375 (11) -0.00125 (12) -15.00625

3.2 Terminating and Non-terminating Decimals

Basically decimals are of the following two types:

- (a) Terminating decimals (b) Non-terminating decimals

3.2.1 Define terminating decimals as decimals having a finite number of digits after the decimal point.

Terminating Decimals:

Let us consider the conversion of some rational numbers

$$\frac{1}{4}, \frac{1}{5}, \frac{1}{2} \text{ and } \frac{3}{8}$$

(i) $\frac{1}{4} = 1 \div 4$ $\begin{array}{r} 0.25 \\ 4 \overline{) 10} \\ \underline{-8} \\ 20 \\ \underline{-20} \\ 0 \end{array}$ Thus $\frac{1}{4} = 0.25$	(ii) $\frac{1}{5} = 1 \div 5$ $\begin{array}{r} 0.2 \\ 5 \overline{) 10} \\ \underline{-10} \\ 0 \end{array}$ Thus $\frac{1}{5} = 0.2$
(iii) $\frac{1}{2} = 1 \div 2$ $\begin{array}{r} 0.5 \\ 2 \overline{) 10} \\ \underline{-10} \\ 0 \end{array}$ Therefore $\frac{1}{2} = 0.5$	(iv) $\frac{3}{8} = 3 \div 8$ $\begin{array}{r} 0.375 \\ 8 \overline{) 30} \\ \underline{-24} \\ 60 \\ \underline{56} \\ 40 \\ \underline{-40} \\ 0 \end{array}$ Therefore $\frac{3}{8} = 0.375$

In above conversion, the process of division is completed and remainder is zero in each case. These rational numbers have been converted into decimals which has limited number of digits after decimal point. These decimals are terminating decimals.

A decimal which contains finite number of digits after decimal point, is called a terminating decimal.

Hence 0.5, 0.25, 0.75, 0.345, 1.7227, 5.023748 etc are all terminating decimals.

3.2.2 Define recurring decimals as non-terminating decimals in which a single digit or a block of digits repeats itself infinite number of times after decimal point. e.g. $\frac{2}{7} = 0.285714285714 \dots$

(b) Non-terminating decimals

Consider the conversion of rational number $\frac{7}{9}$, $\frac{1}{6}$ and $\frac{10}{11}$.

(i) $\frac{7}{9} = 7 \div 9$ $\begin{array}{r} 0.777\dots \\ 9 \overline{) 70} \\ \underline{- 63} \\ 70 \\ \underline{- 63} \\ 70 \\ \underline{- 63} \\ 7 \end{array}$ So, $\frac{7}{9} = 0.777\dots$	(ii) $\frac{1}{6} = 1 \div 6$ $\begin{array}{r} 0.1666\dots \\ 6 \overline{) 10} \\ \underline{- 6} \\ 40 \\ \underline{- 36} \\ 40 \\ \underline{- 36} \\ 4 \end{array}$ So, $\frac{1}{6} = 0.1666\dots$	(iii) $\frac{10}{11} = 10 \div 11$ $\begin{array}{r} 0.909090\dots \\ 11 \overline{) 100} \\ \underline{- 99} \\ 100 \\ \underline{- 99} \\ 100 \\ \underline{- 99} \\ 1 \end{array}$ So, $\frac{10}{11} = 0.909090\dots$
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By converting the given rational numbers to decimals, it is found that

$$(i) \frac{7}{9} = 0.777\dots \quad (ii) \frac{1}{6} = 0.1666\dots \text{ and } (iii) \frac{10}{11} = 0.909090\dots$$

Here the process of division is not complete. These decimals $0.777\dots$, $0.1666\dots$ and $0.909090\dots$ contain infinite number of digits after decimal point. Therefore these decimals are non-terminating.

So,

A decimal which contains infinite number of digits after decimal point is called non-terminating decimal.

In $0.777\dots$ the digit '7' is repeated infinite number of times after decimal.

In $0.1666\dots$ the digit '6' is repeated infinite number of times after decimal.

In $0.90909\dots$ the block '90' is repeated infinite number of times after decimal.

These non-terminating decimals are recurring.

A non-terminating decimal in which a 'single digit' or a 'block of digits' is repeated infinite number of times after decimal point, is called recurring decimal.

Example 1: Convert the given rational numbers as decimals and find whether the decimals so obtained are recurring or not..

$$(i) \frac{2}{3} \quad (ii) \frac{3}{4} \quad (iii) \frac{7}{6} \quad (iv) \frac{9}{10}$$

Solution: We can observe by long division that:

$$(i) \frac{2}{3} = 0.666\dots \text{ and } (iii) \frac{7}{6} = 1.1666\dots$$

Here in both the numbers, digit '6' repeats infinite number of times after decimal, so both are non-terminating and recurring.

$$(ii) \frac{3}{4} = 0.75 \text{ and } (iv) \frac{9}{10} = 0.9$$

Here we obtain zero as remainder, so both are terminating.

Example 2: Convert the given rational numbers $\frac{1}{3}$, $\frac{4}{6}$ and $\frac{2}{7}$ into decimals and find whether the decimals so obtained are recurring or not.

Solution:

(i) $\frac{1}{3} = 1 \div 3$ $\begin{array}{r} 0.333... \\ 3 \overline{) 10} \\ \underline{-9} \\ 10 \\ \underline{-9} \\ 10 \\ \underline{-9} \\ 1 \end{array}$ So, $\frac{1}{3} = 0.333...$ Here 3 is repeated infinite number of times. So, it is a recurring decimal.	(ii) $\frac{4}{6} = 4 \div 6$ $\begin{array}{r} 0.666... \\ 6 \overline{) 40} \\ \underline{-36} \\ 40 \\ \underline{-36} \\ 40 \\ \underline{-36} \\ 4 \end{array}$ So, $\frac{4}{6} = 0.666...$ Here 6 is repeated infinite number times. So, it is a recurring decimal.	(iii) $\frac{2}{7} = 2 \div 7$ $\begin{array}{r} 0.285714... \\ 7 \overline{) 20} \\ \underline{-14} \\ 60 \\ \underline{-56} \\ 40 \\ \underline{-35} \\ 50 \\ \underline{-49} \\ 10 \\ \underline{-7} \\ 30 \\ \underline{-28} \\ 2 \end{array}$ So, $\frac{2}{7}$ is equal to 0.285714285714 ...
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Hence 0.333..., 0.666... and 0.285714... are non-terminating and recurring decimals because the single digit or block of digits are repeated infinite number of times after decimal and the process of division is not complete.

Activity: Identify as terminating and non-terminating.

Rational Number	In decimal form	Terminating	Non-Terminating (recurring)
$\frac{4}{3}$	1.333 ...		Non-Terminating
$\frac{4}{9}$	0.444 ...		
$\frac{5}{4}$	1.25	Terminating	
$\frac{25}{7}$	3.5714285 ...		
$\frac{3}{11}$	0.2727 ...		

EXERCISE 3.2

I. Convert into decimals and identify terminating or non-terminating decimals.

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|----------------------|-----------------------|----------------------|-----------------------|-----------------------|
| (1) $\frac{7}{3}$ | (2) $\frac{11}{6}$ | (3) $-\frac{13}{9}$ | (4) $\frac{15}{7}$ | (5) $\frac{14}{9}$ |
| (6) $-\frac{17}{11}$ | (7) $\frac{19}{12}$ | (8) $\frac{21}{13}$ | (9) $-\frac{23}{14}$ | (10) $-\frac{29}{15}$ |
| (11) $\frac{31}{17}$ | (12) $-\frac{35}{18}$ | (13) $\frac{47}{19}$ | (14) $\frac{35}{23}$ | (15) $-\frac{20}{21}$ |
| (16) $\frac{39}{22}$ | (17) $-\frac{41}{29}$ | (18) $\frac{40}{31}$ | (19) $-\frac{50}{37}$ | (20) $\frac{101}{41}$ |

II. Convert into decimals and identify which is recurring and which is not.

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|---------------------|---------------------|--------------------|---------------------|---------------------|
| (1) $\frac{19}{20}$ | (2) $\frac{17}{6}$ | (3) $\frac{15}{8}$ | (4) $\frac{25}{10}$ | (5) $\frac{2}{11}$ |
| (6) $\frac{10}{3}$ | (7) $\frac{11}{12}$ | (8) $\frac{16}{7}$ | (9) $\frac{2}{13}$ | (10) $\frac{4}{25}$ |

3.2.3 Use the following rule to find whether a given rational number is terminating or not.

RULE: If the denominator of a rational number in standard form has prime factors 2 and 5 or 2, 5 only then that rational number is terminating decimal.

We can identify a rational number whether it is terminating or non-terminating using the above rule without actually dividing.

Consider the following examples.

Rational Number	Division Method	Using Rule	Type of Decimal
$\frac{9}{20}$	$\frac{9}{20} = 9 \div 20$ $= 0.45$	$\frac{9}{20} = \frac{9}{2 \times 2 \times 5}$ Denominator has 2 and 5 as prime factors.	Terminating
$\frac{3}{5}$	$\frac{3}{5} = 3 \div 5$ $= 0.6$	$\frac{3}{5} = \frac{3}{5 \times 1}$ Denominator has 5 as prime factor.	Terminating
$\frac{27}{200}$	$\frac{27}{200} = 27 \div 200$ $= 0.135$	$\frac{27}{200} = \frac{27}{2 \times 2 \times 2 \times 5 \times 5}$ Denominator has 2 and 5 prime factors.	Terminating
$\frac{40}{90}$	$\frac{40}{90} = 40 \div 90$ $= 0.4444 \dots$	$\frac{40}{90} = \frac{2 \times 2 \times 2 \times 5}{2 \times 3 \times 3 \times 5} = \frac{2 \times 2}{3 \times 3}$ Denominator has not 2, 5 as prime factor.	Non-Terminating
$\frac{4}{7}$	$\frac{4}{7} = 0.57142857\dots$	$\frac{4}{7} = \frac{4}{7 \times 1}$ Denominator has not 2, 5 as prime factor.	Non-Terminating
$\frac{27}{18}$	$\frac{27}{18} = 1.5$	$\frac{27}{18} = \frac{3 \times 9}{6 \times 3} = \frac{3 \times 3 \times 3}{2 \times 3 \times 3} = \frac{3}{2}$ Denominator has 2 as prime factor.	Terminating

EXERCISE 3.3

Without actual division, separate the terminating and non-terminating decimals.

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|------------------------|------------------------|-------------------------|------------------------|-------------------------|
| (1) $\frac{12}{25}$ | (2) $\frac{43}{21}$ | (3) $\frac{5}{12}$ | (4) $\frac{17}{20}$ | (5) $\frac{117}{125}$ |
| (6) $\frac{5}{7}$ | (7) $-\frac{9}{40}$ | (8) $\frac{10}{33}$ | (9) $\frac{23}{60}$ | (10) $\frac{40}{35}$ |
| (11) $\frac{55}{75}$ | (12) $-\frac{96}{100}$ | (13) $\frac{101}{125}$ | (14) $\frac{111}{96}$ | (15) $\frac{125}{200}$ |
| (16) $\frac{141}{144}$ | (17) $\frac{200}{201}$ | (18) $-\frac{210}{147}$ | (19) $\frac{372}{400}$ | (20) $-\frac{401}{333}$ |

3.2.4 Express a given rational number as a decimal and indicate whether it is terminating or recurring.

Consider the conversion of following numbers:

- (i) $\frac{7}{20}$ (ii) $2\frac{1}{100}$ (iii) $\frac{1}{7}$ (iv) $\frac{5}{9}$

Solution: (i) $\frac{7}{20} = 0.35$ (ii) $2\frac{1}{100} = 2.01$

As the remainder in both $\frac{7}{20}$ and $2\frac{1}{100}$ is zero.; Therefore these are terminating decimals. Now let us convert $\frac{1}{7}$ and $\frac{5}{9}$ in decimals.

(iii) $\frac{1}{7}$ $\begin{array}{r} 0.1428... \\ 7 \overline{) 10} \\ \underline{- 7} \\ 30 \\ \underline{- 28} \\ 20 \\ \underline{- 14} \\ 60 \\ \underline{- 56} \\ 4 \end{array}$ So, $\frac{1}{7} = 0.1428571...$ and the process of division is not complete.	(iv) $\frac{5}{9}$ $\begin{array}{r} 0.555... \\ 9 \overline{) 50} \\ \underline{- 45} \\ 50 \\ \underline{- 45} \\ 50 \\ \underline{- 45} \\ 5 \end{array}$ So, $\frac{5}{9} = 0.555...$ and the process of division is not complete.
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We observe that rational numbers $\frac{1}{7}$ and $\frac{5}{9}$ are non-terminating and recurring decimals.

EXERCISE 3.4

Express the following rational numbers as decimals. Also indicate whether each is terminating or recurring.

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|------------------------|------------------------|-------------------------|------------------------|------------------------|
| (1) $\frac{5}{14}$ | (2) $-\frac{13}{18}$ | (3) $\frac{19}{20}$ | (4) $-3\frac{21}{22}$ | (5) $6\frac{17}{30}$ |
| (6) $\frac{23}{40}$ | (7) $-\frac{25}{34}$ | (8) $2\frac{15}{26}$ | (9) $3\frac{27}{32}$ | (10) $-1\frac{29}{45}$ |
| (11) $2\frac{31}{38}$ | (12) $\frac{39}{46}$ | (13) $-3\frac{49}{50}$ | (14) $2\frac{45}{48}$ | (15) $\frac{91}{65}$ |
| (16) $-\frac{153}{85}$ | (17) $\frac{266}{171}$ | (18) $-\frac{102}{210}$ | (19) $\frac{111}{228}$ | (20) $\frac{272}{288}$ |

3.3 APPROXIMATE VALUE

Get an appropriate value of a number called rounding off, to a desired number of decimal places.

In calculations we may come across rational numbers which are non-terminating decimals. So, we round off the decimals to some decimal places and then solve our calculations.

RULE: (i) Check the remaining digits.
If the next digit is less than 5 drop it.
(ii) If the next digit is 5 or more than 5, then add 1 to the last digit of the number.

Example: Consider the decimal 0.555... Round off 0.555... to two decimal places.

Solution: We have to round off up to two decimal places.

Step 1: So, we keep two digits of the given decimal same i.e. 0.55

Step 2: As the next digit to be removed after two decimal places is 5.

So, we add '1' to last digit of 0.55
i.e. 5 which will become 6.

Hence the given decimal 0.555... will be rounded off to two decimal places as 0.56

Example 2. Round off 2.333333 to two decimal places, we get: 2.33

Example 3. Round off 4.8888 to two decimal places, we get: 4.89

Activity: Express the following rational numbers by rounding off up to given number of digits.

S. No.	Number	Up to two decimal places	Up to three decimal places	Up to four decimal places
1.	0.05753	0.06	0.058	0.575
2.	2.421875			
3.	0.83333			
4.	1.76794			
5.	0.9916667			
6.	0.96257			
7.	1.61538			
8.	8.08547			
9.	6.98677			
10.	15.47648			

EXERCISE 3.5

I. Convert the following rational numbers as terminating and non-terminating decimals and express up to three places of decimal.

- (1) $\frac{23}{400}$ (2) $\frac{27}{64}$ (3) $\frac{53}{60}$ (4) $-\frac{46}{75}$ (5) $\frac{69}{70}$
- (6) $\frac{77}{80}$ (7) $-\frac{43}{90}$ (8) $\frac{119}{120}$ (9) $\frac{246}{300}$ (10) $-\frac{81}{96}$

II. Identify terminating and non-terminating decimals by division and express them up to two places of decimal.

- (1) $\frac{4}{15}$ (2) $-\frac{11}{12}$ (3) $\frac{8}{9}$ (4) $\frac{22}{7}$ (5) $-\frac{21}{16}$
(6) $\frac{3}{40}$ (7) $-\frac{18}{45}$ (8) $\frac{21}{66}$ (9) $\frac{33}{90}$ (10) $-\frac{81}{101}$

III. Express the following rational numbers as non-terminating or recurring up to three places of decimal.

- (1) $\frac{10}{11}$ (2) $-\frac{41}{90}$ (3) $\frac{21}{13}$ (4) $\frac{17}{19}$ (5) $\frac{9}{7}$
(6) $\frac{15}{23}$ (7) $-\frac{111}{96}$ (8) $\frac{22}{29}$

IV. Round off the following decimals up to two places of decimal.

- (1) 0.4555 (2) 0.3648 (3) 0.6666 (4) 0.1054
(5) 1.9091 (6) 3.2479 (7) 2.0518 (8) 8.3381
(9) 10.1036 (10) 3.9995 (11) 13.0967 (12) 15.8159

V. Round off the following decimals up to three places of decimal.

- (1) 4.51689 (2) 6.7472 (3) 0.0098 (4) 0.9861
(5) 206.4176 (6) 3.4073 (7) 102.99909 (8) 11.1234
(9) 55.12345 (10) 4.10539 (11) 66.3957 (12) 59.9196

REVIEW EXERCISE 3

1. Convert the following decimals into rational numbers.

- (i) 0.325 (ii) 0.75 (iii) 2.5 (iv) 7.75
(v) 0.78 (vi) 1.297 (vii) 2.348

2. Convert the following into decimal fractions and identify terminating and non-terminating fractions.

- (i) $\frac{41}{5}$ (ii) $\frac{111}{12}$ (iii) $\frac{28}{9}$ (iv) $\frac{10}{7}$
(v) $\frac{22}{7}$ (vi) $\frac{21}{16}$ (vii) $\frac{73}{10}$

3. Round off the following up to two-decimal places.

- (i) 14.5828 (ii) 1078287 (iii) 5.7895
(iv) 6.7989 (v) 25.4897

4. Without actual division, separate the terminating and non-terminating decimals.

(i) $\frac{13}{4}$

(ii) $\frac{17}{25}$

(iii) $\frac{80}{3}$

(iv) $\frac{15}{11}$

(v) $\frac{19}{6}$

(vi) $\frac{25}{15}$

(vii) $\frac{22}{7}$

(viii) $\frac{14}{9}$

5. Express the following rational numbers in terminating decimals.

(i) $\frac{21}{100}$

(ii) $\frac{127}{20}$

(iii) $\frac{13}{25}$

(iv) $\frac{31}{50}$

(v) $\frac{145}{1000}$

(vi) $\frac{25}{8}$

(vii) $\frac{23}{6}$

(viii) $\frac{88}{64}$

(ix) $\frac{48}{32}$

6. Convert the following rational numbers in non-terminating decimals and express up to three places of decimal.

(i) $\frac{40}{3}$

(ii) $\frac{20}{7}$

(iii) $\frac{50}{11}$

(iv) $\frac{80}{13}$

(v) $\frac{100}{6}$

(vi) $\frac{240}{22}$

(vii) $\frac{70}{12}$

(viii) $\frac{52}{91}$

7. Round off the following decimals up to three places of decimal.

(i) 5.71679

(ii) 11.60365

(iii) 0.92598

(iv) 3.40855

(v) 0.74396

(vi) 23.15847

8. Answer the following questions.

- (i) What are the terminating and non-terminating decimals?
- (ii) Write the method for converting a decimal into a rational number. Give examples.
- (iii) Which of the non-terminating decimals are called recurring decimal? Give two examples.
- (iv) What is the rule to find whether a given rational number is terminating or not? Give two examples.
- (v) What is meant by the term 'rounding off' in decimals? Give two examples.

9. Fill in the blanks.

- (i) $\frac{1}{2} = 0.5$ which is a _____. (ii) $\frac{1}{3} = 0.333 \dots$ which is a _____.
- (iii) Two parts of a decimal number separated by a dot, are called the _____ and _____.

- (iv) In terminating decimals, division _____ after a finite number of steps.
- (v) A fraction will be terminating decimal if the _____ has 2 or 5 or both as factors of denominator.
- (vi) In recurring decimals, single digit or _____ of digits repeat themselves.

10. Tick (✓) the correct answer.

- (i) To separate a whole number from fractional part in a decimal, we use the symbol.
(a) — (b) . (c) : (d) /
- (ii) If we round off the decimal 3.8461 up to two decimal places, we get:
(a) 3.84 (b) 3.81 (c) 3.86 (d) 3.85
- (iii) A rational number is terminating decimal, if its denominator has no prime factor other than:
(a) 2 and 5 (b) 3 and 6 (c) 4 and 5 (d) 5 and 7
- (iv) When we convert 0.75 into the rational number, we get:
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{3}{4}$ (d) $\frac{1}{7}$

SUMMARY

- Decimals are basically of two types:
(i) Terminating decimals, (ii) Non-terminating decimals.
- Every decimal with finite number of digits after the decimal point is called a terminating decimal.
- A terminating decimal represents a rational number.
- A decimal with infinite number of digits after a decimal point, is called a non-terminating decimal.
- A non-terminating decimal may be recurring or non-recurring.
- A fraction will be terminating decimal if the denominator in standard form has 2 or 5 or both as factors.
- Decimals can be reduced by rounding off the digits upto the desired number of decimal places as under:
(i) Check the remaining digits.
If the next digit, it is less than 5 drop it.
(ii) If the next digit, is 5 or more than 5, then add 1 to the last digit of the number.